

VALUE CHAIN POSITION

Lead Optimisation

PRIMARY
ASO / siRNA safety & activity
PKPD for GalNAc-siRNA

Lead Identification

SECONDARY
OpenOligoLake curation
Hit-level activity prediction

Clinical Pharmacology

SECONDARY
Mechanistic PKPD (OpenNABMPKPD)
Beyond standard popPK/PD

CAPABILITY SNAPSHOT — self-rated + CPDSE read

Competency	Self-rating	CPDSE read
Bioinformatics / Cheminformatics	4 — could teach	Pioneer level. RDKit, oligo featurisers, strong.
Scientific Programming	3 — reliable	Python solid; conda envs; not containerised.
Data Management	3 — reliable	PostgreSQL + Git. Manual extraction steps remain.
ML / Predictive Modelling	3 — reliable	XGBoost + model cards. No UQ on safety models.
Reproducible Research	2 — ad hoc	Git used. PKPD not reproducible cross-machine.
Statistical Analysis	2 — ad hoc	Bayesian partial. Formal testing inconsistent.
Data Visualisation	2 — ad hoc	Functional plots. No standard templates.

! CORRECTION ZONE

Add sticky notes here for anything
the pipeline got wrong or missed.

Value chain wrong? Move it.

Rating too high / too low?

Data gap we missed?

Constraint we do not know about?

FACILITATOR: Ask "which rating is most wrong, and in which direction?" — not "do you agree?" | When challenged, say "put that on the board."

Surface DATA GAP callouts explicitly: "We could not verify whether Jonas has Docker competence — is that right?"



KEY FRAME — Structural vs. Technical gaps

STRUCTURAL

Affects everything regardless of method.
Reproducibility, data management, workflow design, team knowledge.
Fixing one improves all research threads simultaneously.

TECHNICAL

Affects one method or capability.
Missing tools, domain knowledge, specific methods.
Fixing one improves that thread only.

FINDING 1

Uncertainty quantification is now standard for safety-critical ML

Groups publishing oligo safety prediction in 2024-25 routinely include conformal prediction (MAPIE library) alongside point predictions. A model that says "I don't know" on out-of-distribution compounds is more publishable — and more useful — than one that confidently mispredicts.

QUESTION FOR THE ROOM:

Is this a choice or a gap for your group?

FINDING 2

Your PKPD work sits outside the standard framework — that is a feature, not a bug

Standard Clinical Pharmacology tools (NONMEM, Monolix) are calibrated for popPK/PD in clinical settings. Your mechanistic compartment modelling for NABMs is biophysically more detailed. There is no established benchmark — which means you are building it.

QUESTION FOR THE ROOM:

Does this feel like strength or exposure right now?

FINDING 3

Transfer learning is the tractable path on data sparsity — but sequence matters

Pre-trained molecular models (ChemBERTa-77M) fine-tuned on small datasets are the field approach to your sparsity problem. But you need uncertainty quantification first — otherwise you cannot tell if a fine-tuned model is better calibrated or just more accurate on training data.

QUESTION FOR THE ROOM:

Where does this sit in your mental roadmap?

CONSULTANT FIELD NOTES — What changed? What needs updating in the gap assessment?

FACILITATOR: Present findings 1-3 as provocations, not verdicts. Finding 2 (PKPD) should land hardest with Morten — if it generates pushback, it is working.

Finding 1 connects directly to Block 3 gaps. Finding 3 connects to Block 4 sequencing. Use those bridges explicitly.



YOUR JOB:

1. Which gap is most **WRONG**? Drag it to the challenge zone.

2. Which gap is **MISSING**? Add a sticky note.

3. Which gap is most **URGENT** for you? Dot vote (3 dots each).

Everyone votes — not just Morten. Master students: your urgency read matters.

STRUCTURAL GAPS

(affect everything — fix one, improve all)

STRUCTURAL

PKPD pipeline non-reproducibility

Reproducible Research

HighLow effortActive

STRUCTURAL

OpenOligoLake extraction inconsistency

Data Management

HighMedium effortActive

STRUCTURAL

Emma thesis transition risk

Data Management

HighMedium effortNear-term

STRUCTURAL

Inconsistent statistical rigour

Statistical Analysis

MediumMedium effortNear-term

TECHNICAL GAPS

(affect one method — fix one, improve that thread)

TECHNICAL

No uncertainty quantification on ML models

ML / Predictive Modelling

HighMedium effortActive

TECHNICAL

No transfer learning strategy

ML / Predictive Modelling

HighHigh effortNear-term

TECHNICAL

Model cards not retroactively applied

Reproducible Research

MediumLow effortNear-term

TECHNICAL

No standardised visualisation templates

Data Visualisation

LowLow effortNear-term

! CHALLENGE ZONE

Drag cards here that you think are wrong, overstated, or missing.

Add a sticky note explaining why.

MISSING GAPS

Add sticky notes for gaps

we did not identify:

FACILITATOR:

Emma succession risk (card 3) will create silence — name it directly: "Emma, this one is about you. Is the 12-month timeline accurate?"

Read the dot voting pattern, not just the total. If Morten and master students diverge, surface it: "You are seeing urgency differently — why?"



THE QUESTION: "Which of these would you actually do in the next 4 weeks if we were not in the room?"

That question is the filter. If the answer is "none of them," something went wrong earlier. Assign owner + date before moving to strategic.

QUICK WINS — 0-3 months

QUICK WIN

Containerise PKPD workflow

Gap: PKPD non-reproducibility

Done by: _____

Owner: _____

Effort

2-3 person-weeks

QUICK WIN

Structured extraction template

Gap: OligoLake extraction inconsistency

Done by: _____

Owner: _____

Effort

3 person-weeks

QUICK WIN

Retroactive model cards

Gap: Model cards not applied

Done by: _____

Owner: _____

Effort

2-3 person-days

QUICK WIN

Shared visualisation style sheet

Gap: No standard visualisation

Done by: _____

Owner: _____

Effort

2 person-days

STRATEGIC — 3-18 months (agree direction today, plan detail later)

Emma succession plan

By month 3 start. Must start before Emma enters thesis mode.

!

Uncertainty quantification on ML models

MAPIE conformal prediction. Must precede any Roche model sharing.

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Standardise Bayesian PKPD protocol

After PKPD containerisation quick win is done.

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Transfer learning exploration

After UQ implemented. ChemBERTa-77M starting point.

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SEQUENCING — order matters



COMMITMENTS — fill before leaving

Three things. Three owners. Three dates. Right now.

#	Commitment	Owner	Done by
1			
2			

WORKSHOP 2: 6-8 weeks from today. CPDSE will follow up in 2 weeks to check quick win progress. Brief handover happens NOW — last page of the strategy brief is the commitment table.

FACILITATOR: When assigning owners, ask "who will ACTUALLY DO this, not who is responsible?" | Morten should not own more than one quick win.

Emma succession verbal commitment from Morten is non-negotiable — do not leave without it. | Then hand over the brief.